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A TRACKING CONTROL NEURAL NETWORK FOR THE KNOWN AFFINE IN THE INPUTS DISCRETE-TIME NONLINEAR SYSTEMS

Keywords: Difference equation, discrete-time, nonlinear system, tracking control, variable structure.

Abstract

A neural network (NN) tracking control for discrete-time nonlinear systems with the known internal dynamics is designed. The control is described by the control equation based on the tracking equation. These equations have variable structure. To minimize the tracking error, the control uses a difference between the system state trajectory and reference trajectory. A corresponding functional block-diagram of the controlled system is presented. Principles of software and hardware implementation of the control are discussed. The convergence time can be changed by varying the learning rate parameter. The control does not need an offline learning phase and has architecture of moderate complexity. Computer simulations of the controller operation are provided.

Introduction

A tracking control problem can be formulated as designing such a system whose output approximately repeats a given time-varying reference trajectory that is assumed to be predefined. Tracking control systems are widely used in telecommunications, automobiles, ships, aircrafts, robotics, biomedical engineering, and in many other applications. For instance, problems such as the fulfilling ship movement along a required route, performing aircraft flight or carrying out manipulator arm movement are related to solving tracking control tasks [1].

Different approaches of tracking control for nonlinear systems were proposed [2 - 9]. For example, a neural network (NN) structure to solve the optimal tracking control problem of nonlinear discrete time-varying systems in online mode is presented in [5]. A multiple-model approach combined with adaptive self-organizing map NN is used to detect changes in the system dynamics. The number of sub-models is determined adaptively and grows once a mismatch between the stored sub-models and the new data is detected. Each sub-model contributes into the value function based on a responsibility signal obtained by the NN. Policy iteration and value iteration algorithms are suggested to find the optimal control for the partially unknown nonlinear discrete time-varying systems in online mode. In the majority of the existing methods, a convergence of tracking error to zero is theoretically guaranteed during an infinite range of time. Moreover, numerous modern approaches of tracking control need a training phase of cost function approximation. This leads to enormous rising complexity of the control system. Besides, the procedure of such training can be too time consuming. As a result, the time complexity and computational complexity of the tracking control problem solution are considerably increased.

A NN tracking control for continuous-time nonlinear systems with the known internal dynamics was presented in [8]. The control is described by the control equation based on tracking equation. Both these equations have variable structure. To minimize the tracking error, the control uses a difference between the system state trajectory and reference trajectory. A functional block-diagram of the controlled system is given. Principles of software and hardware implementation of the control are discussed. The control provides finite convergence time of the system state trajectories to steady states and decreasing tracking errors. The convergence time can be changed by varying the learning rate parameter. The control does not need an off-line learning phase and has architecture of moderate complexity. Computer simulations of the controller operation are presented.

This paper is related to extending the method presented in [8] in the case of designing a tracking control NN of discrete-time for affine in inputs nonlinear systems with the known

internal dynamics. In this case, the designed network is described by control equation and tracking equation that have variable structure. The network employs a difference between the system state and reference for tracking error minimization. The network is accurate and has less computation and hardware implementation complexity comparatively to other competitors. Computer simulations confirming theoretical statements and illustrating the controller efficiency are given.

1. A DISCRETE-TIME TRACKING CONTROL NEURAL NETWORK

Discrete-time NNs compared to their continuous-time analogs are capable of providing more accurate, robust and reliable processing data and signals, especially in the presence of noises and errors [10]. Such networks can be implemented in a digital hardware that has a high miniaturization level and energy efficiency. Using Euler approximation, the discrete-time tracking control problem of time-invariant, affine in the input nonlinear dynamic system can be given by [9]:

$$x(k+1) = x(k) + \tau(f(x(k)) + g(x(k))u(k)), \quad x(1) = x_1, \quad (1)$$

where $x(k) \in \mathbb{R}^n$ is the state variable, $f(x(k)) \in \mathbb{R}^n$ is the nonlinear function of system internal dynamics, $g(x(k)) \in \mathbb{R}^{n \times m}$ is the nonlinear function of system coupling dynamics, $u(k) \in \mathbb{R}^m$ is the control, $\tau = t(k+1) - t(k)$ is the time step, $k=1, 2, \dots$

We suppose that $f(0)=0$ and the system is stabilizable on a set $\mathcal{V} \in \mathbb{R}^m$ that includes the origin, i.e., there exists such a function $u(k) \in U$ of discrete-time control that the system is asymptotically stabilized on $\mathcal{V}$. The functions of system internal dynamics $f(x(k))$ and $g(x(k))$ are presumably known for any $k \in [1, k_{max}]$, $k_{max} < \infty$. We also suppose that the system is observable and controllable, i.e., $g(x(k)) \neq 0$. Otherwise if $g(x(k))=0$, the system is uncontrolled and can be given by

$$x(k+1) = x(k) + \tau f(x(k)), \quad x(1) = x_1. \quad (2)$$

The real-valued cost function associated with (1) can be presented as follows:

$$C(x) = \sum_{k=1}^{\infty} \Phi(e(k), u(k)), \quad (3)$$

where $e(k) = x(k) - r(k)$ is the tracking error, $r(k) \in \mathbb{R}^n$ is the reference trajectory, $\Phi(e, u) \in \mathbb{R}^n$ is the positive definite function. The tracking control problem is to determine a control $u(k)$ such that, starting from any finite initial state $-\infty < x(1) < \infty$, as $k \rightarrow \infty$, (3) is minimized, i. e., optimal value function

$$C^*(x) = \min_u \sum_{k=1}^{\infty} \Phi(e(k), u(k)), \quad (4)$$

while the state $x(k)$ remains bounded. The equation (4) means that starting from any initial value, (3) should converge to its minimum, i. e., the system should be stable. In order to solve the tracking control problem of the known affine nonlinear system stated above, it is necessary to design a control such that $x(k)$ tracks a given varied reference trajectory $r(k)$.

We design the cost function (3) as follows:

$$C(x) = |e|. \quad (5)$$

Then, the optimal value function (4) is given by

$$C^*(x) = \min_u |e|, \quad (6)$$

where control u is defined as indicated above while the state x remains bounded.

Since the cost function C in (5) is convex, this guarantees that it has a unique global minimum. Its minimizer $C^* = 0$ can be found using a gradient descent method [10]. For this purpose, C can be minimized by the following gradient system:

$$\Delta e = -\nabla C(e) = \frac{\partial C(e)}{\partial e} = \left[\frac{\partial C}{\partial e_1}, \dots, \frac{\partial C}{\partial e_n} \right]^T, \quad (7)$$

where $\Delta e = e(k+1) - e(k)$ and initial condition $e(1) = e_1$. Considering (5), the system (7) can be transformed to the following associated difference equation with the variable structure that describes the closed-loop tracking error dynamics:

$$e(k+1) = e(k) - \beta \operatorname{sgn}(e(k)), \quad (8)$$

where $\beta = \operatorname{diag}(\beta_1, \dots, \beta_n)$ is the diagonal matrix of learning rate parameters,

$$\operatorname{sgn}(e) = \begin{cases} 1, & \text{if } e > 0; \\ 0, & \text{if } e = 0; \\ -1, & \text{if } e < 0 \end{cases} \quad (9)$$

is the signum activation function [11] of tracking error $e = x - r$.

The state equation (8) can be represented in the equivalent deployed form as follows:

$$x(k+1) = x(k) - \beta \operatorname{sgn}(x(k) - r(k)) + \Delta r(k), \quad (10)$$

where

$$\operatorname{sgn}(x(k) - r(k)) = \begin{cases} 1, & \text{if } x(k) > r(k); \\ 0, & \text{if } x(k) = r(k); \\ -1, & \text{if } x(k) < r(k), \end{cases} \quad (11)$$

$\Delta r(k) = r(k+1) - r(k)$. The tracking equation (10) describes a discrete-time state variable trajectory of the designed system.

The control $u(k)$ can be determined from (1) considering (10) by the following control equation:

$$u(k) = g^{-1}(x(k))(x(k+1) - x(k) - f(x(k))) = g^{-1}(x(k))(-\beta \operatorname{sgn}(x(k) - r(k)) + \Delta r(k) - f(x(k))). \quad (12)$$

In the case if $g(x(k)) = 0$, the control equation (12) can be given by

$$u(k) = \begin{cases} g^{-1}(x(k))(-\beta \operatorname{sgn}(x(k) - r(k)) + \Delta r(k) - f(x(k))) & \text{if } g(x(k)) \neq 0; \\ 0, & \text{if } g(x(k)) = 0. \end{cases} \quad (13)$$

The difference equation (1) and control equation (12) (or (13)) describe the full closed loop dynamics of the nonlinear system. Substitution $u(k)$ (12) in (1) gives the tracking equation (10). As one can see in (10), $x(k)$ and $\Delta x(k) = x(k+1) - x(k)$ can be derived by solving the difference equation for each time step $k, k=1, 2, \dots$. Based on determined $x(k)$ and $\Delta x(k)$, $u(k)$ can be obtained using (12) by solving the system of linear algebraic equations. A solution of such a system can be found in online mode by corresponding NN implemented in software or in digital hardware [11]. Then, the desired discrete-time state variable trajectory $x(k)$ of the controlled system is found by numerical integration of the difference equation (1) using known values of $x(1) = x_1$, $f(x)$, $g(x)$, and $u(k)$.

The main steps of the algorithm of optimal tracking control for $l=1, \dots, n$ time-invariant, affine in the input nonlinear system with the known internal dynamics can be presented as follows:

Step 1. Set values of β , control time range T , τ , $k=1$, and $x(1)$.

Step 2. Set/obtain/measure $r(k)$, $r(k+1)$ and compute $\Delta r(k) = r(k+1) - r(k)$.

Step 3. Compute $f(x(k))$ and $g(x(k))$.

Step 4. Compute $u(k)$ by (12).

Step 5. Compute $x(k+1)$ using (1).

Step 6. If $k < T/\tau + 1$, set $k=k+1$, go to step 2; otherwise end.

Notice that the difference equation (10) includes the signum activation function (11). Moreover, this equation describes supervised learning with learning rate parameter β that can be used for changing convergence rate of state variable trajectory $x(k)$ to reference trajectory $r(k)$. In addition, the control $u(k)$ is determined by (12) on each time step $k, k=1, 2, \dots$ by solving the system of linear algebraic equations that can be done in online mode using the corresponding NN. Since the activation function and supervised learning are the basic

characteristics of artificial NNs [10, 11], we treat the discrete-time tracking controller described by the tracking equation (10) and control equation (12) as a tracking control NN.

2. FUNCTIONAL BLOCK-DIAGRAM OF THE CONTROLLED NONLINEAR SYSTEM

A functional block-diagram of the controlled nonlinear system described by (1) and (12) is presented in Fig. 1. The diagram includes the blocks of discrete-time summation $+$, *sign* activation function S , amplification β , numerical differentiation Δ , nonlinear functions $f(x)$ and $g(x)$, NN solver of system of linear algebraic equations LAESS, reference r , state variable $x(k)$ with initial value $x(1)$, and control input $u(k)$. As one can see in Fig. 1, the NN consists of $3n$ adders, n controlled switches, n amplifiers, n digital differentiators, n blocks of nonlinear functions $f(x)$, $n \times m$ blocks of nonlinear functions $g(x)$, and n NN solvers of a systems of linear algebraic equations.

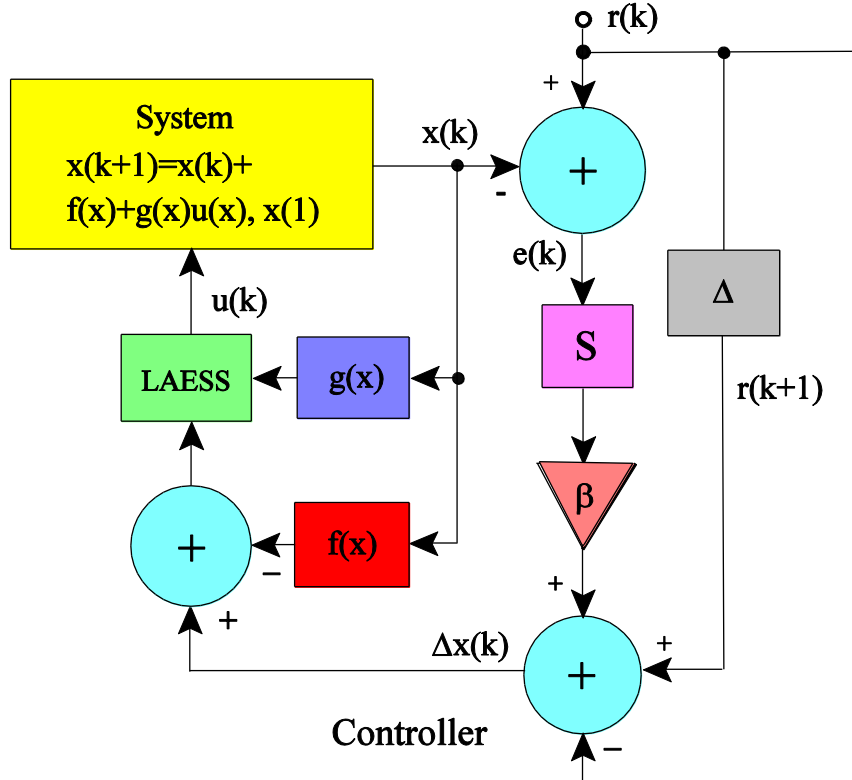


Fig. 1. The functional block-diagram of the discrete-time controlled system described by (1) and (12).

The network presented in Fig. 1 has the structure of moderate complexity compared to other competitors. For instance, actor NN and critic NN are required for optimal tracking control of nonlinear discrete-time systems described in [4].

The network shown in Fig. 1 can be implemented in software on a PC or workstation with no special hardware. Software implementation of the controller can be chosen if accurate, simple and flexible realization of changing the application is necessary. For this purpose, one can use, for instance, Compute Unified Device Architecture (CUDA) [12] or the parallel computing toolbox of Matlab. Moreover, using the power of modern Graphics Processing Units (GPU) makes it possible to implement the network by GPU code which significantly speeds up its operation.

In the case of software implementation of the network, its operation speed can be insufficient to satisfy the requirements of real-time. Moreover, in this case the level of miniaturization or/and energy efficiency of the network can be not sufficient to meet the requirements of practical applications. Therefore, the network can be implemented in a digital hardware suitable for parallel operation in real-time, if high operation speed, high integration density, and energy efficient realization are needed. Note that the hardware implementation of

the network can be efficiently simulated using Cadence, LabVIEW or other appropriate software [13].

Notice that the network does not need resetting in the case if its state or reference is varied, a suitable digital control network, and does not require to spend extra processing time on this operation. This additionally reduces its complexity, speeds up the network operation and makes it possible to use it in real-time. Therefore, digital hardware implementation of the network is expected to have high speed, moderate complexity, compactness, and energy efficiency.

3. COMPUTER SIMULATIONS OF THE NETWORK

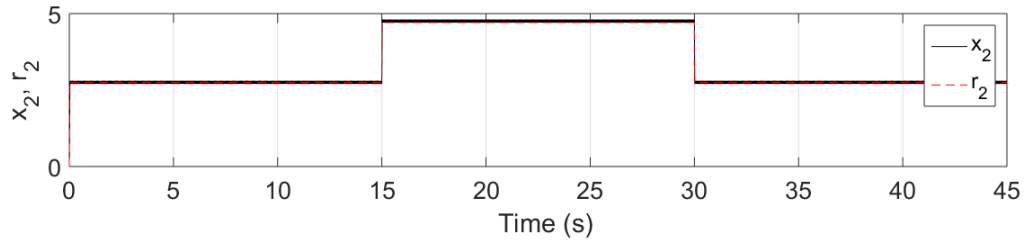
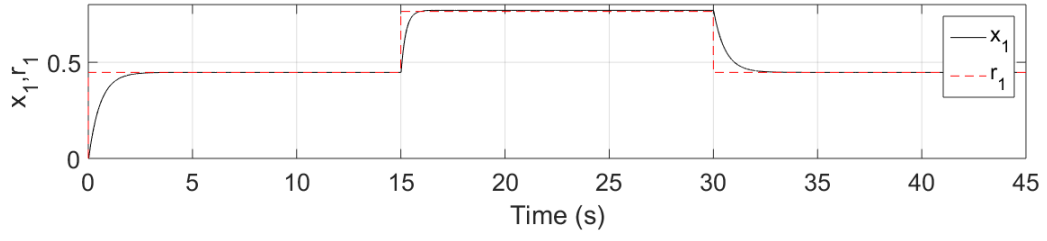
Let us demonstrate an operation of the discrete-time affine in the input controlled nonlinear system with the known internal dynamics described by the difference equation (1) and control equation (12). For this purpose, consider the following computer simulation example. We apply the approach described above to the optimal tracking control of nonlinear cooling continuous stirred tank reactor adopted from [2, 14, 15]. The dynamics of the reactor can be described in normalized form by the nonlinear difference equation (1), where

$$f = \begin{bmatrix} x_1(k) + \tau \left(-\alpha x_1(k) + D(1 - x_1(k)) e^{\frac{x_2(k)}{1 + \frac{x_2(k)}{\gamma}}} \right) \\ x_2(k) + \tau \left(-\alpha x_2(k) + BD(1 - x_1(k)) e^{\frac{x_2(k)}{1 + \frac{x_2(k)}{\gamma}}} - \lambda x_2(k) \right) \end{bmatrix}, \quad g = \begin{bmatrix} 0 \\ \lambda \end{bmatrix}, \quad \alpha = 1.0, \quad \lambda = 0.3, \quad \gamma = 20.0,$$

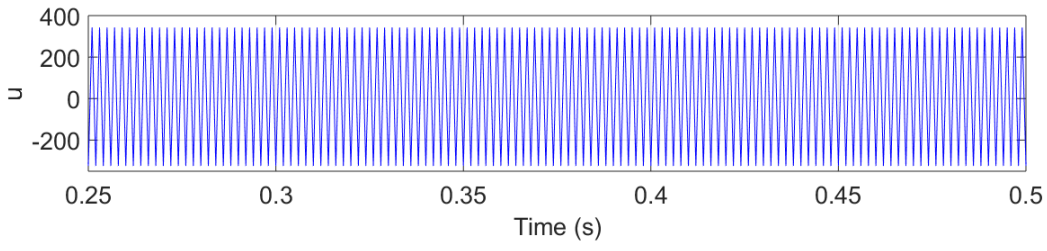
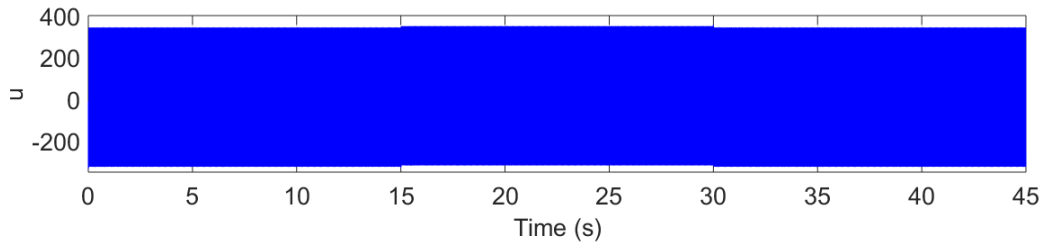
$B = 1.0$, $D = 0.072$, $\beta = 100$, $\tau = 0.001$ s, $x_1(1) = x_2(1) = 0$. The purpose is to track the reference

$$\text{trajectories } r_1(t) = \begin{cases} 0.4472, & \text{if } 0 \leq t \leq 15; \\ 0.7646, & \text{if } 15 < t < 30; \\ 0.4472, & \text{if } 30 \leq t < 45 \end{cases} \text{ and } r_2(t) = \begin{cases} 2.7520, & \text{if } 0 \leq t \leq 15; \\ 4.7052, & \text{if } 15 < t < 30; \\ 2.7520, & \text{if } 30 \leq t < 45. \end{cases}$$

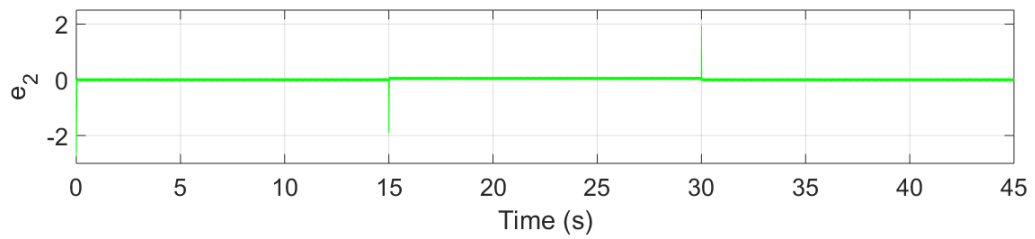
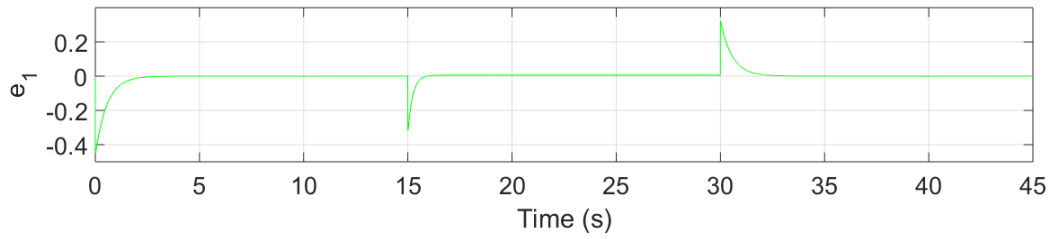
The trajectories of state variables $x_1(t)$, $x_2(t)$ obtained based on the difference equation (1) and control equation (12) and references $r_1(t)$, $r_2(t)$ are presented in Fig. 2 (a). Fig. 2 (b) shows the dynamics of control $u(t)$. Fig. 2 (c) demonstrates the trajectories of tracking errors $e_1(t)$, $e_2(t)$. Fig. 2 (a) – Fig. 2 (c) show that the state variable trajectory $x_1(t)$ tracks the reference trajectory $r_1(t)$ with the accuracy close to that obtained in [2] and the state variable trajectory $x_2(t)$ tracks the reference trajectory $r_2(t)$ with higher accuracy comparatively to one indicated in [2]. The control $u(t)$ and tracking errors $e_1(t)$ and $e_2(t)$ are varied in limited ranges. However, the method presented in [2] needs solving minimization problems at each iteration using recurrent general projection NN to obtain optimal control. For this purpose, ordinary nonlinear differential equation is solved. Our approach is related to obtaining optimal control by solving nonlinear algebraic control equation (12). This considerably reduces computational complexity and corresponding hardware implementation complexity of the optimal tracking control problem solution.



(a)



(b)



(c)

Fig. 2: (a) trajectories of states $x_1(t)$, $x_2(t)$ and references $r_1(t)$, $r_2(t)$; (b) trajectory of control $u(t)$; (c) trajectories of tracking errors $e_1(t)$, $e_2(t)$.

4. CONCLUSIONS

A tracking control neural network for discrete-time nonlinear dynamic systems with known internal dynamics described by difference equation with variable structure and by output equation have been presented. The mathematical model and corresponding functional block-diagram of the network have been given. The network is accurate and simpler than other competitive analogs. The controller can be implemented in software if accurate, simple, and flexible realization of changing the application is necessary. The network can be also implemented in a digital hardware suitable for parallel operation in real-time, if high operation speed, high integration density, and energy efficient realization is needed. A performance of the network has been demonstrated by simulation of optimal tracking control of nonlinear cooling continuous stirred tank reactor. It has been shown that using the network considerably reduces computational complexity and corresponding hardware implementation complexity of optimal tracking control.

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